Nova Nexus

System Introduction

System Application

This system represents a prototype for a pedometer.

- Would be most applicable for general purpose health monitoring.
- Designed to be accessible

General Description

The main purpose of this system is to monitors steps, also includes other functions to facilitate this. Features 3 buttons each having a specific function, and 3 LEDs for easy general pace monitoring.

Team Introduction

DYLAN PURBRICK	ANIKA ROTH	BRANDON RAYAN	LACHLAN SCOTT ROGERS
ROLE:	**ROLE:**	**ROLE:**	**ROLE:**
Team Lead	PCB Designer	Programming Lead	CAD and Assembly Lead
SUBCIRCUIT:	**SUBCIRCUIT:**	**SUBCIRCUIT:**	**SUBCIRCUIT:**
Power Circuit	Top Level Schematic	Filter Circuit	Accelerometer Circuit
SUBROUTINE:	**SUBROUTINE:**	**SUBROUTINE:**	**SUBROUTINE:**
Step Tracking	Walking Pace Identification	Calibration	Self-Test Routine

Communication

1 **Microsoft Teams Chat**

2 **In Person Collaboration**

3 **Trello Board**

1

Also the issues and solutions needs to be done for embedded software Brandon Rayan and enclosure Lachlan Scott Rogers as well as anything else (Dylan Purbrick if you know what else please say)

Brandon Rayan 9:16 AM

Yep I'll be alternating between working on pfms and ems today so I'll work on the testing and implementation

1

We also need slides for the preso if they haven't been done yet

Anika Roth 9:17 AM

Yeh it's been done it's on the teams page under files in final presentation or smth

Dylan has also made a start on the datasheet so that is in there as well

9:23 AM

Also needed is just a general description and features of the pedometer, also just updating the case and its evolution might be needed.
and then we can adjust our group evaluation on the final page but making sure everything else is in order first is best.

1

Brandon Rayan 11:34 AM

hey Anika Roth did you do any code for step identification?

if so did you push it?

Brandon Rayan 1:19 PM

Everyone Ok worked on the code and it seems that everything works as intended.

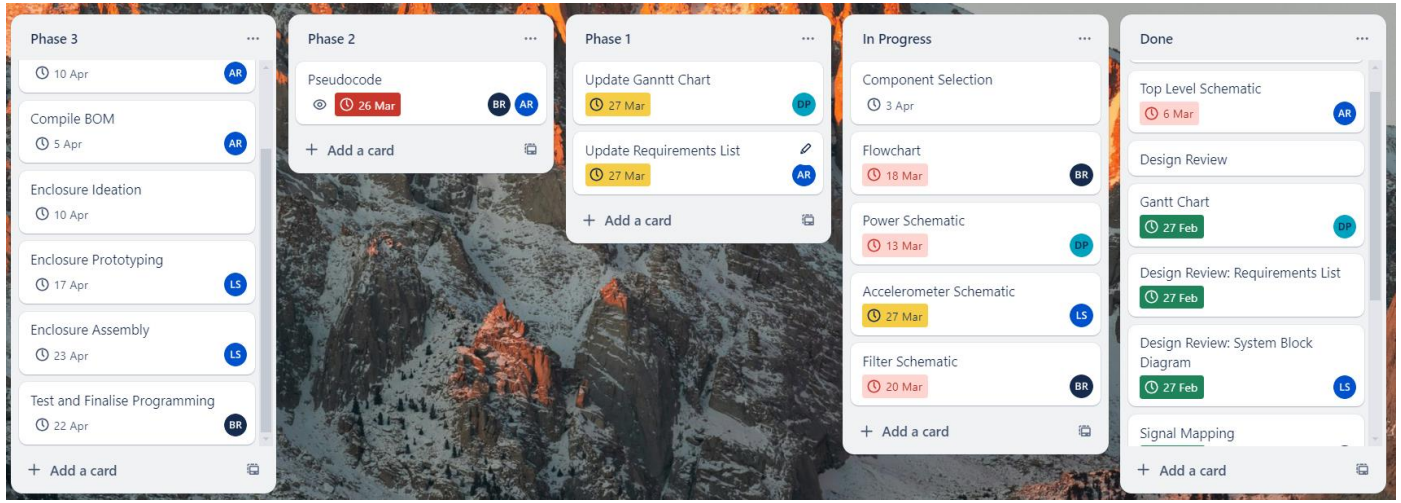
Lachlan Scott Rogers I have the variables that you wanted in the main file and have commented them specifically

So you are good to finish up the routine

Lachlan Scott Rogers 1:21 PM

Yeah finishing mine should be easy now, might just need a hand testing it

3



Requirements List

PROGRAMMING

Requirement	Value
Min Walking Pace	>0 m/s
Max Walking Pace	10 m/s
LCD should convey Step Count	
Program should be written in the Arduino Programming Language	
Walking Pace Tolerance	0.005m/s
X-Axis output if Vs = 3V	-1.08 g
Y-Axis output if Vs = 3V	+1.08 g
Z-Axis output if Vs = 3V	+1.83 g

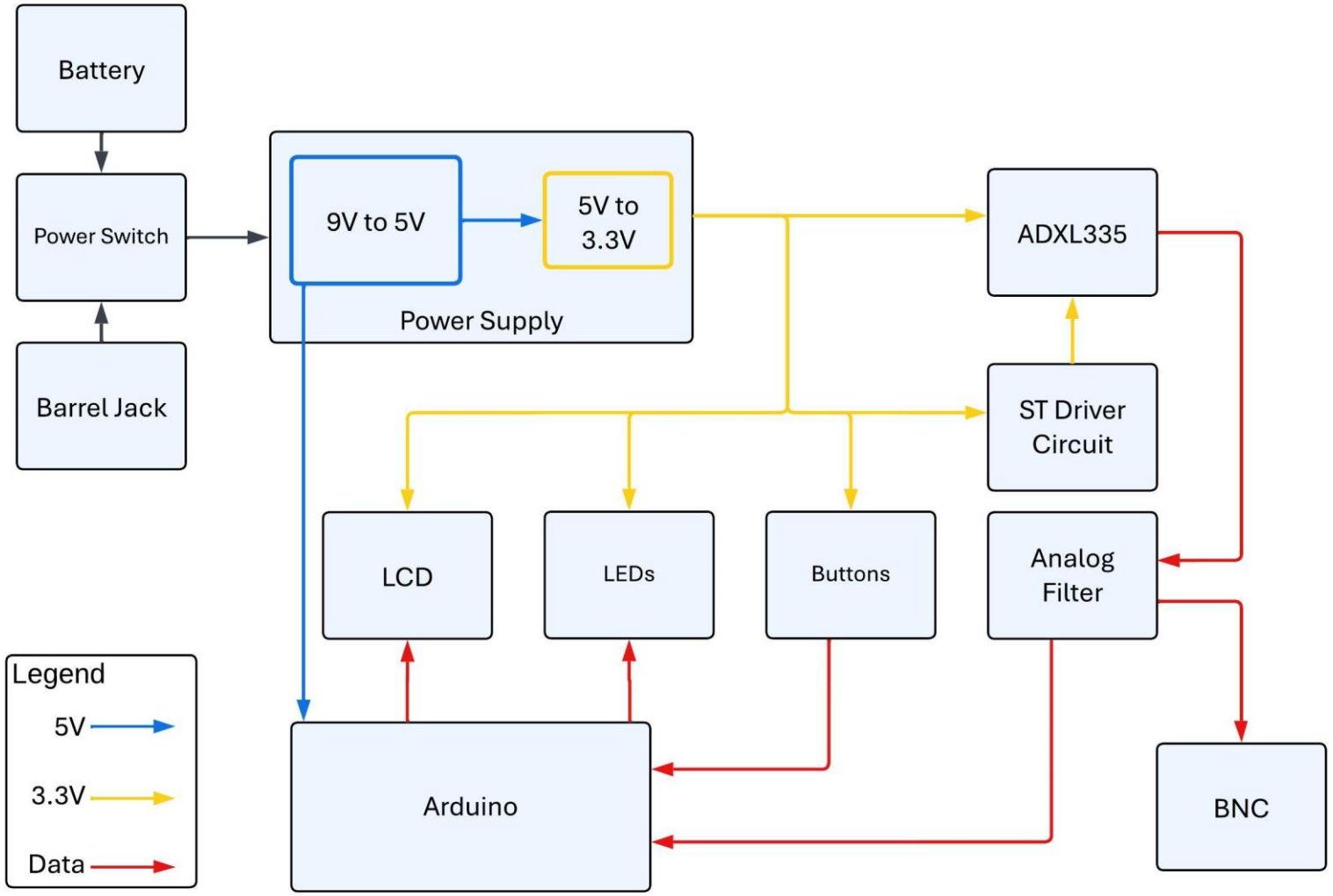
PCB

Requirement	Value
Max Size	102.9mm x 80.1mm
Min No. of Test Points	9
Min Power Output on PCB	+ 3.3V
Should physically connect to pins on Arduino Uno R3	
LCD and LED should convey meaningful outputs to the User	
User should be able to interact with system	

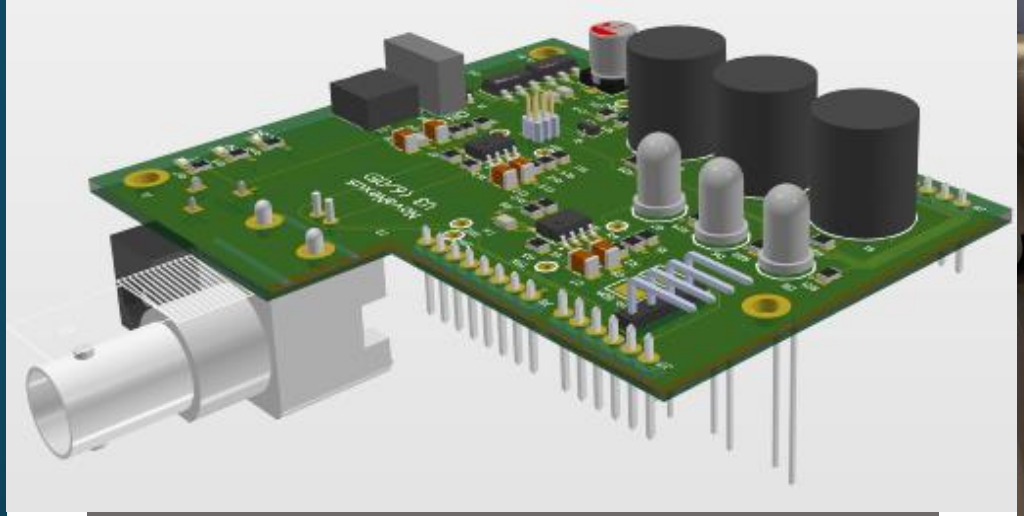
ENCLOSURE

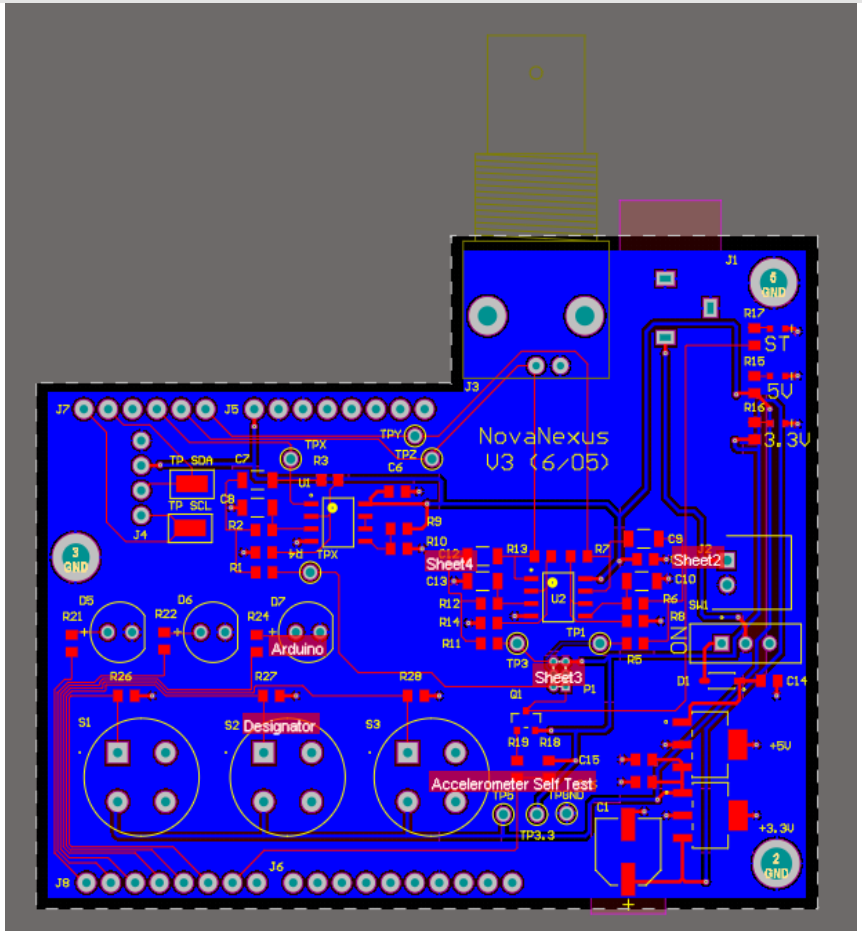
Requirement	Value
Peripherals should be clearly visible through enclosure	
Max Size	150mm x 150 mm x150mm
Max Weight	50 g
Should be able to mount to the human body (shoulder/wrist) and experimental interface	

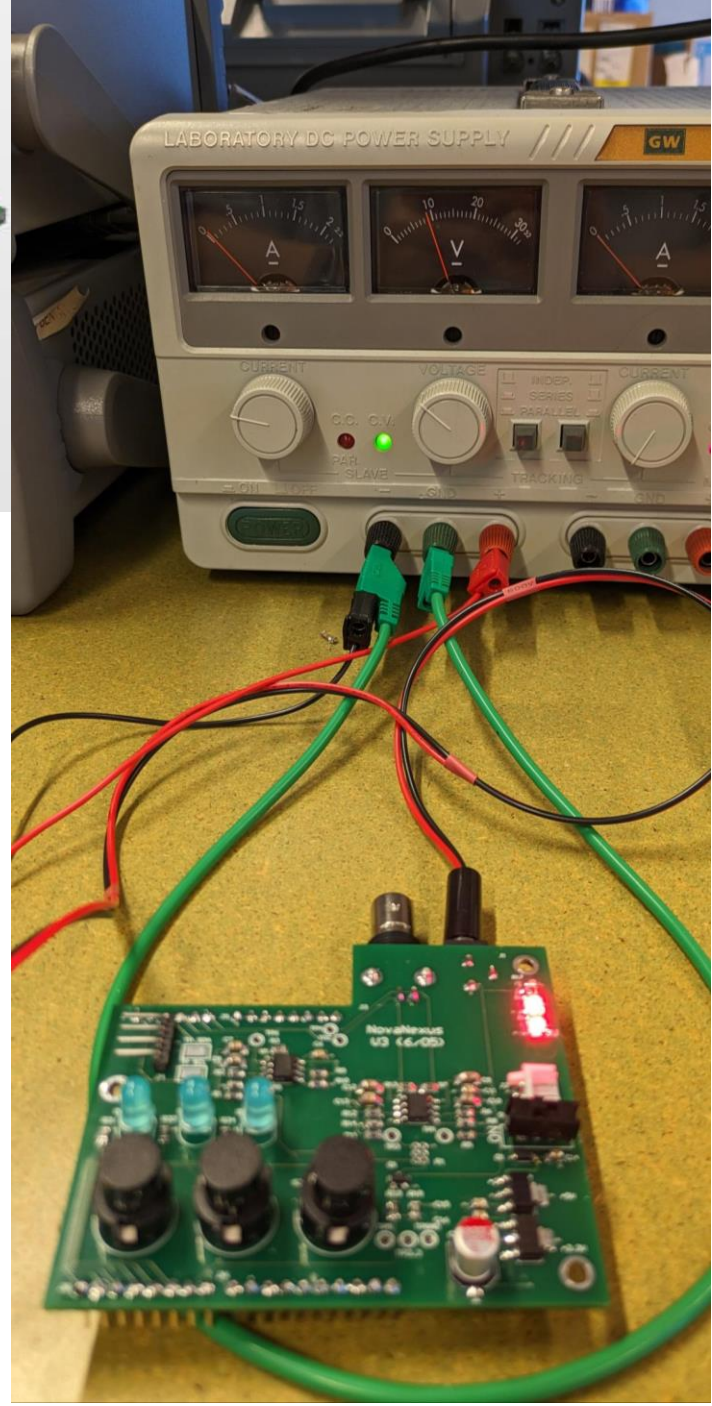
System Block Diagram



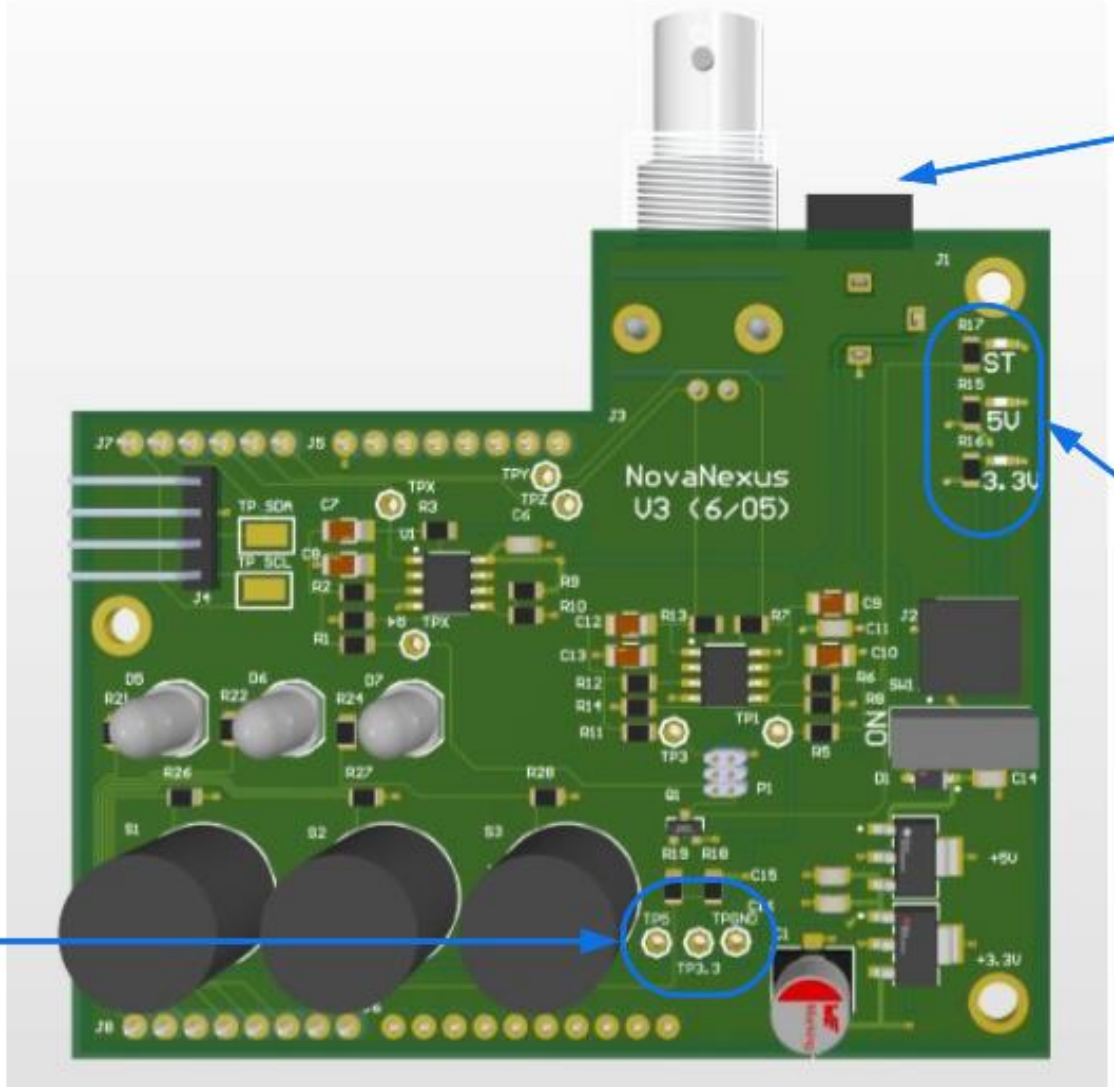
The PCB




NovaNexus
V3 (6/05)
Sheet4
Sheet2
Sheet3
Arduino
Designator
Accelerometer Self Test
ST
5V
3.3V


LABORATORY DC POWER SUPPLY
GW

PCB Innovative Features



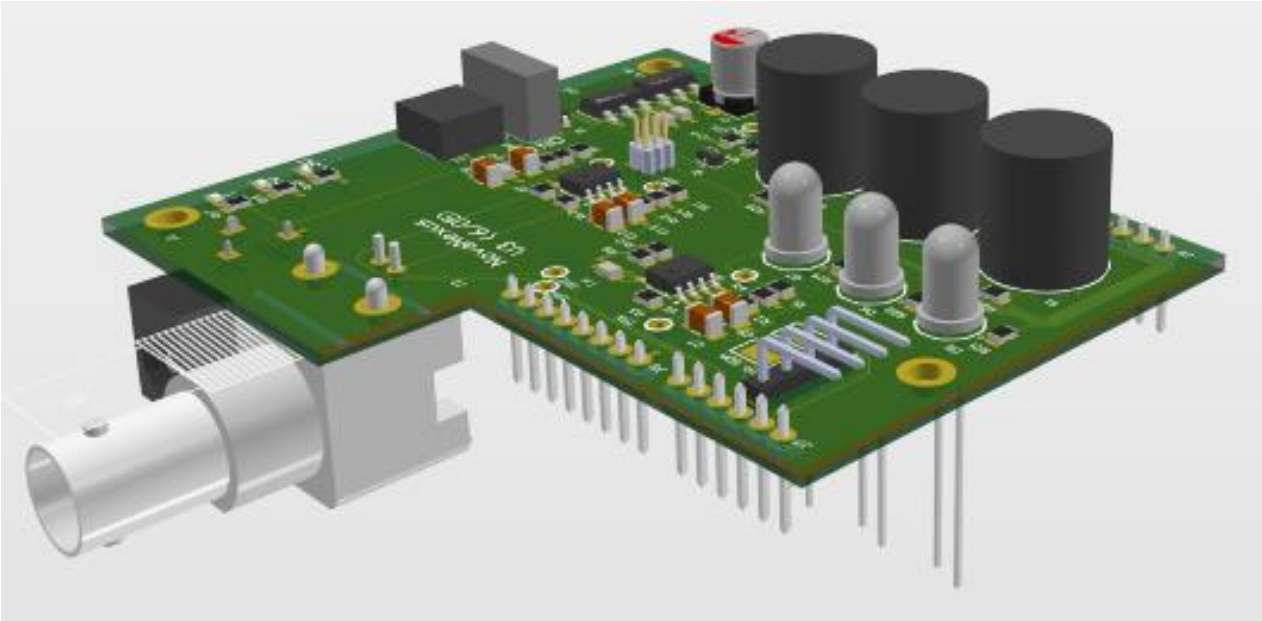
BNC and Power Barrell are sticking out and on the bottom

Localised SMD Indicator Lights

Localised Power Test Points

User Consideration

Choosing buttons that can be secured solidly on the PCB but whilst allowing for user interaction



Power Barrel and BNC positioned away from user and Buttons closest to user

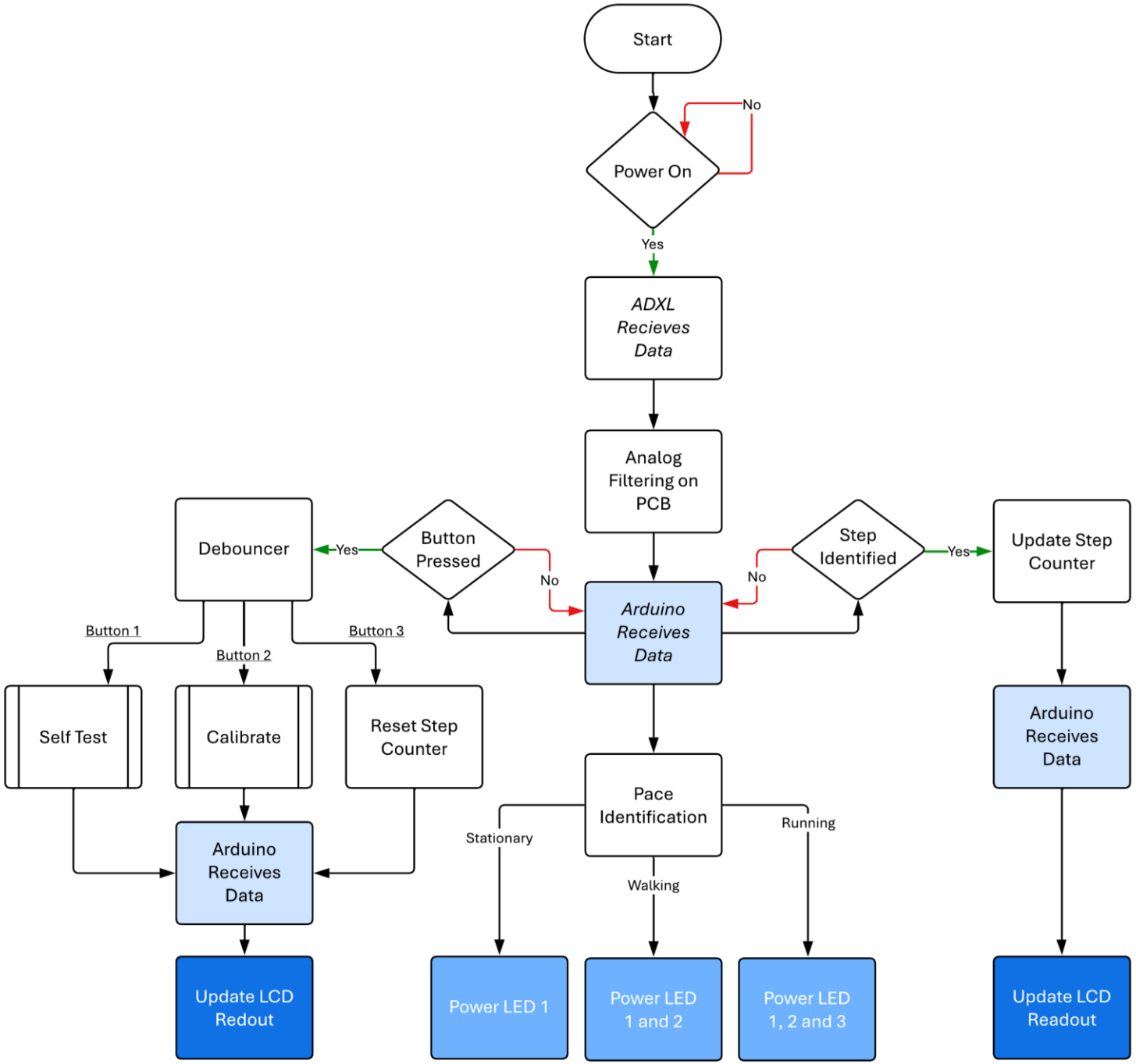
LEDs positioned to be easily viewed without being a hazard

Issues and Resolutions: PCB

Issue 1: Accelerometer Connector	Issue 2: Filter Schematic	Issue 3: Opamp Positioning
Discovered when checking and comparing to pin outs on accelerometer	Power indicator lights on PCB were not turning on and power source was being current limited.	Power indicator lights on PCB were not turning on and power source was being current limited.
Due to looking at the pin allocation of the header part without referencing the actual accelerometer pin outs the wiring did not work.	The filter schematic had the wrong circuit section connected to the positive and negatives.	The filter had been soldered on incorrectly and so the pins weren't connected properly
Option 1: Hack the board to make it work Option 2: Revise the part footprint and PCB design and order a new board	Option 1: Hack the board Option 2: Revise schematic, PCB design and order new board	Option 1: Unsolder and resolder the Opamps into correct position

Honourable Mentions	
	• Components would have hit Arduino • Some signal wires would cross over • Melting Switches

Functional Flowchart



Embedded System Programming

ADXL335 Test Protocol

Static Acceleration Calibration

Step Counting

Walking pace Identification

Programming Functions

ADXL335 Test Protocol

Function is used to ensure that the Accelerometer remains functional. This is to be activated via a button and output to the LCD when the protocol is completed

Static Acceleration Calibration

Function that eliminates static gravitational force acting on the accelerometer in all directions (i.e gravity). This is initiated via a button and outputs to the LCD when the calibration is complete

Step Counting

Function that identifies when a step occurs and increments when a step occurs. A button is used to reset the counter and the number of steps are outputted to the LCD

Pace Identification

Function uses the frequency and magnitude of the steps to determine if the user is stationary, walking or running. The blue LED's indicate what pace the user is at

Issues and Resolutions: Embedded Software

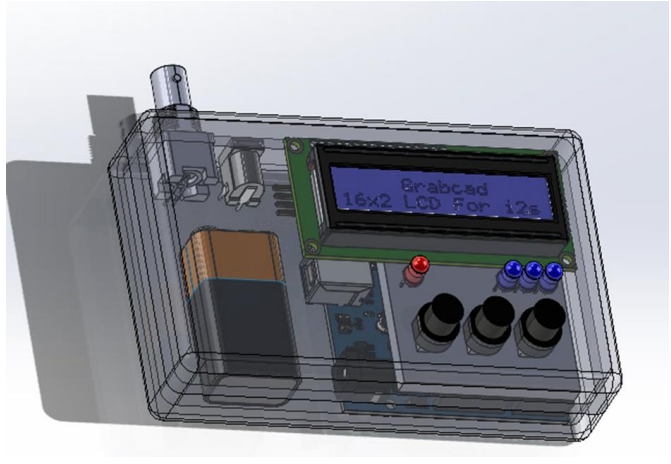
Issue 1: Sampling
The Step Identification had accuracy issues and would miss steps so we talked to Michael
In the sampling function samples were created from the average of 10 values. This added an extra layer of filtering that interfered with the output used in calculations
Option 1: Create a new sampling function that only used one value. Option 2: Continue Averaging but smaller sample size

Issue 2: Walking vs Running
The Stride identification had accuracy issues and would easily jump between walking and running
The conditional bounds for walking and running were very close together and calibration value did very minimal to fix this issue
Option 1: Verify Calibration code Option 2: Do more experiments using phyphox to determine new values for each set pace

Issue 3: LCD Overlap
LCD outputs were overlapping with each other
The button code used for the Self test and calibration would not enter a different state and thus ran in conjunction with the code in the main loop
Option 1: Rewrite the button logic to incorporate a state machine in conjunction with debouncers

Honourable Mentions	
	• Use of Serial Monitor interferes with LEDS • Z-axis influences the average too much

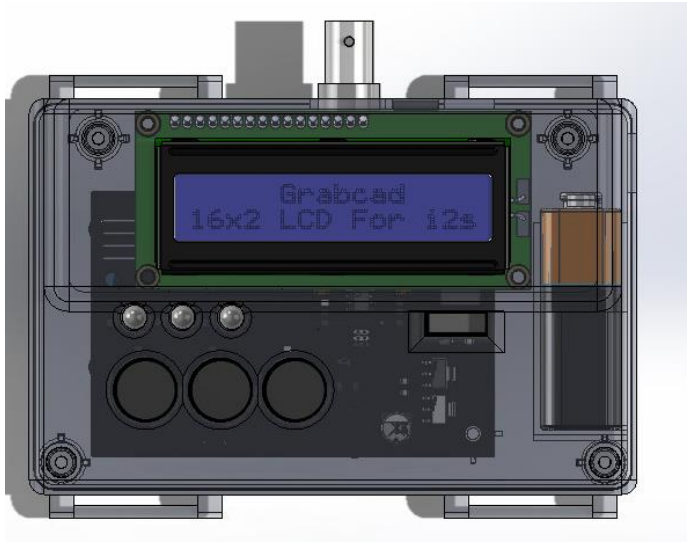
Enclosure Evolution

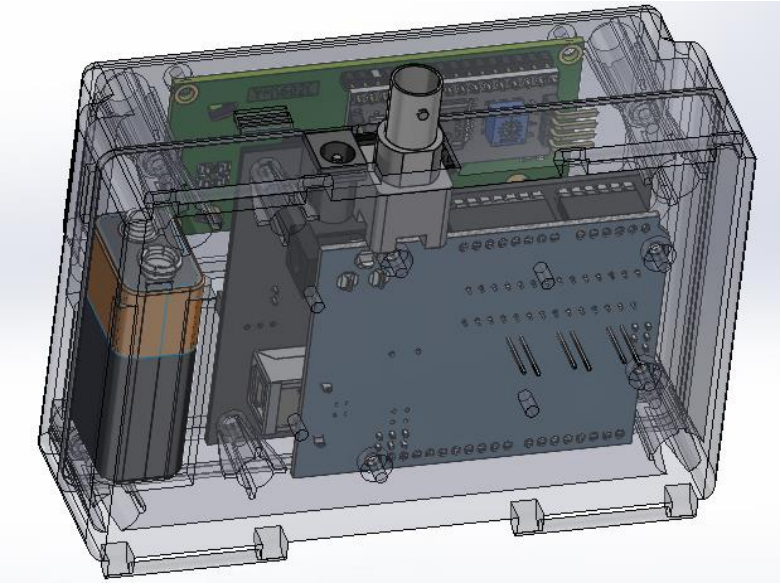


V1	V2	V3	V4
Basic initial enclosure design	MCAD Codesigner allowing PCB in project	Further refinement of design	Further refinement of design
Visual indication of space needed for components	PCB informed location of cutouts for: • LED's • Buttons • Switch • BNC connector • Connector	Support posts	Height of lower lid section raised to avoid collisions
Planning the layout	Raised LCD section	Fastening points	Lug & Pin design finalized
Introduction of LCD cutout	Orientation of Battery	Lip & Groove	PCB location moved
		Lug & Pin designs tested	
		Viewing slot for indicator LED's	
		Battery flipped to other side of enclosure	

Final Enclosure

CAD Model



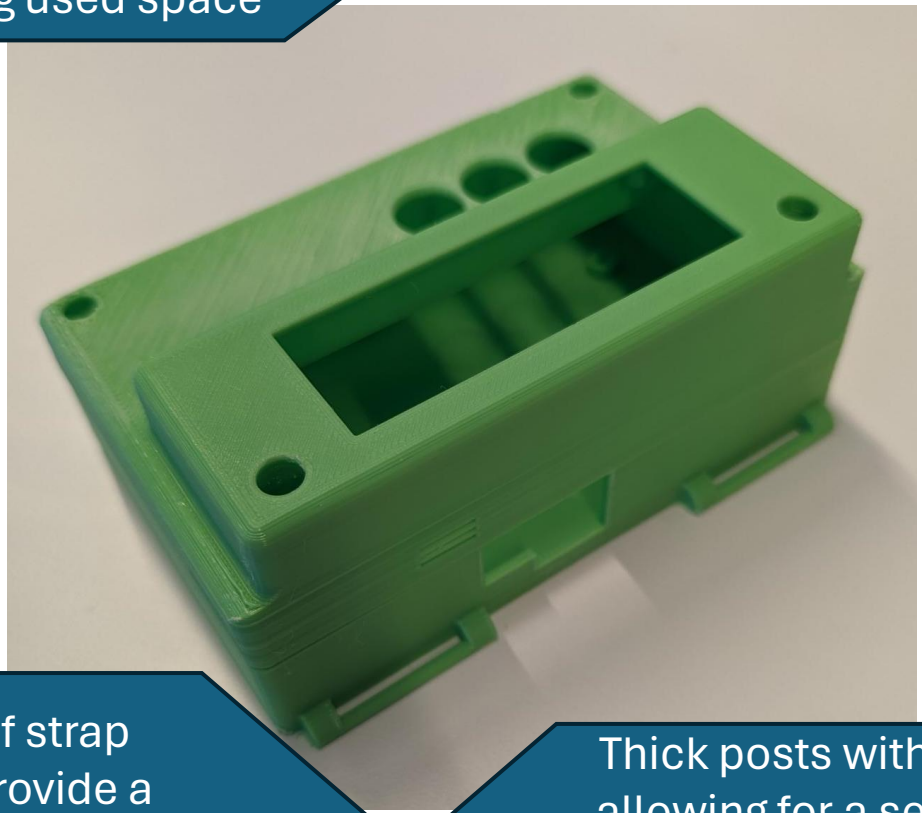


3D Printed Enclosure

Enclosure Innovative and Creative features

Finding the perfect height for the buttons to be accessible while still accommodating ADXL

LCD has been offset to allow easy viewing by the user and minimizing used space



Two pairs of strap holders to provide a secure and adjustable fit between users

Thick posts with fins allowing for a secure and durable fit between top and bottom

Issues and Resolutions: Enclosure

Issue 1: Design Intent
The Enclosure would not properly code and would hit with components needed for the operation of the Step Tracker
Some components, specifically the ADXL and connecting wires, were not accounted for in the CAD model and so on the actual print there was not enough space to accommodate them
Option 1: Create a new model with adjustments and reprint Option 2: Create cutouts for ADXL or decreasing connector length

Issue 2: Tolerance
Fitment between top and bottom segments very tight to the point of a substitute snap fit and screw holes were much too small.
CAD model had not accounted for the tolerances of the 3D printer. Different printers would result in different tolerances.
Option 1: Create a new model with accounting for tolerances based on one specific 3D printer to use

Issue 3: Stand-Offs
Stand-offs were positioned too close to the edge and were susceptible to breakage
Design oversight for the main intent of the stand-offs and expectations in terms of tolerances.
Option 1: Adding fins to improve durability Option 2: Moving towards the center of case

Honourable Mentions	
	• Switch and button interactability • Securing Battery • Print Bed Size and Printer Error

User Experience

- **Buttons**
  - Used to interact with the sub routines of the system
  - Tactile and ease of use in nature
- **LEDs**
  - Indicate types of strides that the user is in
  - Bright and operates as a progress bar
- **LCD**
  - Outputs key information for the user relating to the function of the step tracker
    - No. of Steps
    - Stride state
    - Progress of sub routines

Group Evaluation

MARKING CRITERIA	LEVEL ACHIEVED
Presentation Quality & Delivery	Expert
Team Profile, Collaboration & Communication	Expert
System Overview & Specifications	Proficient
PCB Innovation & Creativity	Proficient
Enclosure Innovation & Creativity	Proficient
Embedded Innovation & Creativity	Proficient
Evaluation, Feedback & Reflection	Proficient